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ORIGINAL ARTICLE

Exploring user engagement and perceptions towards smart energy applications in Norway: Insights from a text mining analysis of user reviews

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Abstract Smart energy applications can be powerful tools for two-way communication between energy providers and end-users. These apps offer real-time usage insights and present consumption data through intuitive and visually appealing interfaces. In parallel, these platforms allow users to provide ratings, reviews, and comments directly within or via app provision platforms, thereby helping energy providers improve their overall services. Long-term engagement with such energy apps could benefit sustainable energy consumption practices, as such digital platforms can empower users to make informed energy-efficient choices. Nonetheless, there is a lack of a systematic approach to analysing users' engagement towards such smart energy apps and users' opinions about their practical functionalities and features. This study examines how users interact with smart energy apps, assess their functionalities, and share their opinions through ratings, reviews, and comments. It offers unique insights into end-user perspectives through advanced text-mining techniques incorporating word analysis, sentiment analysis, topic modelling, and exploratory qualitative analysis. By analysing over 3,000 publicly available user-generated reviews, the research identifies patterns such as monitoring energy consumption, managing appliances, and integrating smart home features like EV charging. The findings highlight diverse motivations for app engagement, extending beyond cost savings to include empowerment, convenience, and alignment with sustainability goals. While users express predominantly positive sentiments about the app's intuitive design, real-time feedback, and gamification features, reviews also reveal areas for improvement, emphasizing the importance of user-centred design in enhancing app usability and functionality. The findings provide actionable insights for improving app design, advancing smart energy solutions, and informing energy efficiency and conservation initiatives.

Keywords Sustainable household · User engagement · Smart energy apps · Norway · Technology adoption · Sentiment analysis · Topic modelling

Abbreviations

App	Application
BERT	Bidirectional Encoder Representations from the Transformers
COVID-19	Coronavirus Disease 2019
EV	Electric Vehicle
iOS	IPhone Operating System
kWh	Kilowatt hour
LDA	Latent Dirichlet allocation
NLTK	Natural Language Toolkit
NLP	Natural Language Processing
PV	Photovoltaics

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TF-IDF	Term frequency-inverse document frequency
TWh	Terawatt hour

Introduction

The residential energy sector is witnessing a surge in the availability and adoption of feedback-based digital solutions, including in-home energy displays, web portals and smartphone-based energy apps. Among these, energy apps are gaining attention as tools to facilitate household energy management and foster engagement (Valor et al., 2019). Research suggests that feedback mechanisms provided through such apps can encourage user engagement, enhance energy awareness, and potentially facilitate sustainable energy practices (Veskioja et al., 2022). The feedback from the users through these apps can also guide energy providers in refining their operations and services, including whether the apps' features continue to engage users and motivate them to monitor their energy consumption patterns regularly.

Despite the growing adoption of these technologies, recent systematic reviews have highlighted both their transformative potential and inherent limitations, underscoring the need for more comprehensive exploration of their long-term impacts and scalability (Chatzigeorgiou & Andreou, 2021; Wemyss et al., 2023). A critical shortcoming in existing literature lies in the underrepresentation of the complexity of user interactions with smart energy technologies and the nuanced ways these interactions mediate user perceptions and engagement (Christensen et al., 2020; Skjølsvold et al., 2018; Standal et al., 2023). While empirical studies have explored the role of various feedback mediums in influencing energy practices (Hargreaves et al., 2010), there remains a research gap in understanding user interactions with energy apps as a medium to facilitate sustainability (Valor et al., 2019).

Moreover, existing studies predominantly rely on theoretical frameworks or pilot-scale interventions rather than leveraging real-world data from active users (Paneru & Tarigan, 2023). The two main issues in conducting this type of study are the lack of data regarding users' engagement with smart energy apps and the limited knowledge to analyse such data. This study addresses this gap by investigating user interactions with the Tibber energy app, a digital platform widely adopted by Norwegian households, to explore how such technologies mediate engagement, perceptions (Paneru et al., 2023) and energy practices (Paneru & Tarigan, 2023; Simonsen et al., 2022). Norway's position as a global leader in deploying smart energy technologies, characterised by its unique socio-technical context, provides an optimal environment for this inquiry (Korsnes & Throndsen, 2021).

This study employs a mixed-methods approach incorporating word analysis, sentiment analysis, topic modelling, and exploratory qualitative analysis to comprehensively understand user experiences, perceptions, and engagement with the app. It analyses public user reviews and identifies recurring themes such as user motivations, perceived app value, and reported challenges. These findings contribute to the broader discourse on the digitalization of energy services, uncovering general trends in user engagement and identifying opportunities to enhance app design and functionality to promote energy-conscious practices. Overall, this study is guided by the following research question:

RQ How do user interactions with energy applications shape engagement, perceptions, and energy management, and what implications do these interactions have for designing user-centric digital energy solutions?

The research is structured as follows: Sect. "Background" contextualizes the relevance of app-based energy feedback and user reviews in Norway. Sect. "Methodological Approach" details the methodological framework employed in the study. Sect. "Analysis and Findings" presents the analysis and key findings, while Sect. "Discussions" discusses the findings. Sect. "Limitations and Further Research" outlines the study's limitations and proposes directions for future research. Finally, Sect. "Conclusion" summarizes the findings and broader implications for the field.

Background

App-based energy feedback and user engagement

App-based energy feedback has emerged as a promising digital solution to enhance user awareness and engagement with energy usage. A systematic review by Chatzigeorgiou and Andreou, (2021) examined feedback research for residential energy behaviours

through mobile and web interfaces, highlighting the need for more comprehensive studies on these solutions' long-term impacts and scalability.

Smart energy apps offer context-specific feedback that allows users to track and reflect on their consumption patterns (Hermsen et al., 2016). This feedback can assist users in exploring management strategies, yet the extent to which these apps shape sustainable energy practices within households remains under-researched and still limited to pilot-level projects (Attour et al., 2020; Geelen et al., 2019; Murugesan et al., 2015; Valor et al., 2019). Further, Wemyss et al. (2023) emphasise the importance of moving beyond the initial excitement around these technologies to understand better the conditions under which they function as social innovations contributing to low-carbon transitions. This perspective underscores the need to study energy apps as mediators of user engagement, providing insight into how digital interfaces influence interactions with energy data and inform decision-making.

The COVID-19 pandemic has further amplified interest in digital solutions that support sustainable lifestyles as environmental awareness became a more salient priority for consumers (World Economic Forum, 2020). Lockdowns and remote work increased reliance on digital tools for managing daily activities, including energy consumption (Gunaratna & Aslam, 2022). This shift created opportunities for digital energy solutions, such as smart energy apps, to play a greater role in household energy management. Verplanken and Wood (2006) highlight that smart energy technologies, including energy apps, can encourage reflective energy use by providing actionable feedback. Similarly, Sanguinetti et al. (2018) emphasise that user interaction with these technologies can enhance awareness of energy-efficient practices. However, Huijer, (2013) observes the effectiveness of these tools and examines how well they integrate with users' existing practices and routines.

Furthermore, Chalal et al., (2022) conducted a systematic review of energy eco-feedback systems, highlighting the importance of effective visualisation techniques in enhancing user engagement and comprehension of energy consumption patterns. Recent work by Mosca et al., (2024) maps the landscape of digital interventions to promote sustainability, calling for future studies on effective digital interventions. This study builds on this literature by focusing on how users engage with digital energy platforms, using app reviews as a lens to explore perceptions, evaluations, and points of interaction.

Relevance of user reviews

Analysing user reviews for smart energy apps can offer valuable insights into users' perceptions and interactions with these technologies. Particularly, such reviews can highlight how users engage with app features, their awareness of energy feedback, and their perceptions of the app's utility in supporting informed energy decisions. This approach aligns with the growing recognition of energy big data's potential to enhance efficiency by analysing user dimensions (Zhou & Yang, 2016).

As digital technology advances, numerous online platforms such as Twitter (officially known now as X), Instagram, Facebook and other specialised online forums have emerged, which serve as a virtual space for individuals to exchange their opinions, perspectives, and concerns, and they are becoming increasingly popular and influential in providing a real-time record of public sentiment (Arlt et al., 2018; Fronzetti Colladon, 2018). Research on such social media content can offer a deeper understanding of the social perception, social contexts of specific policies, and potential solutions to bridge existing gaps (Li et al., 2019). Fronzetti Colladon, (2018) highlights that spontaneous consumer expressions can provide valuable insights into product features and user experience when analysed without creating a sense of surveillance. In this context, user-generated reviews on app-provision platforms such as Google Play Store and Apple App Store offer a useful means of understanding users' perceptions and feedback on the apps they use (Cambria et al., 2016). These reviews serve as a valuable source of information, and systematically analysing the vast amount of user-generated content can help reveal evolving attitudes towards smart solutions, particularly in relation to household energy use (Genc-Nayebi & Abran, 2017; Mariani & Borghi, 2022). The increasing availability of these reviews has also given researchers an exceptional opportunity to collect natural data without direct interaction with subjects (Heinonen & Medberg, 2018). This approach is straightforward and cost-effective, and prior studies

have highlighted the benefits of using online user content to analyse users' perceptions and reported experiences (Buchanan et al., 2014; Huang et al., 2022; Lee et al., 2020; Serrano et al., 2021).

Research leveraging user-generated reviews has been valuable in sectors such as health (Dennison et al., 2013; Zečević et al., 2021), tourism (Serrano et al., 2021), sharing economy (Lee et al., 2020), green consumption (Huang et al., 2022) and customer service (Li et al., 2023). These studies demonstrate the potential of text-mining techniques to uncover users' sentiments, motivations, and perceived benefits associated with digital tools. However, within the energy sector, limited research applies such methods to analyse user reviews of household energy apps at the population level.

Although user reviews do not fully capture detailed household energy practices, they offer valuable insights into user engagement with the app. By examining users' reported experiences, motivations, and perceived benefits, researchers can understand users' interactions and potential interest in energy management practices within households.

Norwegian context

Household consumption and energy choices are significant in the sustainable energy transition (Dubois et al., 2019). In this study, energy consumption refers to electricity usage, reflecting its predominance in Norwegian households. The residential sector is critical for promoting sustainable energy practices, as buildings accounted for 34% of Norway's energy consumption in 2020, with 55% attributed to residential buildings (Norsk Industri, 2021). Norwegian households exhibit exceptionally high electricity consumption, averaging around 16,000 kWh per household in 2016 (Statistics Norway, 2018). This high consumption is influenced by several factors, including the widespread use of electric heating, colder climates, relatively low electricity prices, and the prevalence of larger homes equipped with multiple appliances (energifakta, 2024). This high consumption makes the residential sector critical for promoting sustainable energy practices.

Despite a gradual decline in average household size from 2.2 in 2013 to 2.11 in 2023 (Statistics Norway, 2023), residential energy consumption is projected to increase due to the growing integration of power-intensive appliances and smart technologies into households (Norsk Industri, 2021). This increase is occurring alongside broader transformations in Norway's energy system, including decentralisation, the adoption of energy storage technologies, the rise of renewable energy prosumers, and increasing climate-induced challenges such as fluctuating hydropower reservoir levels (energifakta, 2022; IFE, 2024; Paneru & Tarigan, 2023). This underscores the need for effective supply–demand management (Askeland et al., 2021; Goda et al., 2018; Viseth, 2022).

Norwegian policymakers have set ambitious goals to reduce energy consumption in existing buildings by 10 TWh by 2030, compared to the 2016 levels, with an estimated total energy-saving potential of approximately 12 TWh in the private building sector (Simonsen et al., 2022; Wachenfeldt, 2009). Achieving these targets requires a multifaceted approach, including enhanced energy awareness, smarter consumption management and changes in household energy practices.

The Norwegian government's energy policy documents underscore the importance of home automation systems in enhancing energy efficiency and demand flexibility supported by regulations to drive adoption (GoN, 2021, 2022). This aligns with the vision of an ideal smart and sustainable household electricity consumption (Skjølsvold et al., 2018).

Norway's electricity market operates across five distinct price zones (NO1-NO5), designed to manage grid congestion and address regional supply–demand imbalances by providing efficient price signals to consumers and producers based on local conditions (NVE-RME, 2024). The widespread adoption of smart meters in Norwegian households has further enabled users to interact with digital energy platforms, such as energy apps. These apps are becoming increasingly significant tools for consumers to engage with their electricity usage.

The growing availability of user reviews provides a valuable dataset for exploring how users perceive and engage with these technologies (Paneru et al., 2023). By analysing user interactions with energy apps, this study positions Norway as a particularly relevant context for understanding the role of digital platforms within a well-established energy system. This research contributes to broader discussions on the digitalization of energy services, highlighting their potential to empower end-users and advance sustainable energy transitions.

Methodological approach

Table 1 outlines the analytical methods employed in this study to examine user engagement with smart energy apps. The following sections elaborate on the methodological steps, beginning with data collection, dataset overview, and analytical framework. This study employs a mixed-methods approach, integrating both quantitative and qualitative analyses. The objectives are to explore users' awareness, perceptions, sentiment, key discussion topics, and the relationships between these topics. Additionally, qualitative analysis provides deeper insights into user concerns that may not be fully captured by quantitative data.

Data collection

Selection of app

Tibber was selected as the case study for this research based on several key criteria. First, it has gathered a solid user base, with the highest number of downloads among energy apps in Norway, with almost 100 k + downloads on the Android platform (Google Playstore, 2022) and over 9.5 k + downloads on the iOS (Appstore, 2022), indicating strong market penetration and user adoption (Paneru & Tarigan, 2023). Second, it has extensive user reviews available on Android and iOS platforms, providing a wealth of publicly accessible data for analysis. Third, it offers distinctive functionalities such as social comparison and broad compatibility with smart home devices, aligning with the study's objectives.

Although there are multiple apps in the Norwegian market, we focused solely on Tibber for this study due to its distinctive features, market position, and suitability for our research. Tibber ranks 7th on the 2023 Digital Innovation Index, the highest among Norwegian energy companies (NHH, 2023). It also incorporates more advanced visualisation techniques and gamification elements than similar apps, making it particularly suitable for studying user perceptions and household energy practices (Paneru & Tarigan, 2023). Although the Tibber app is only available to its customers, households in Norway can freely choose and switch their electricity supplier (energifakta, 2022). Further, Tibber is an energy supplier focused on smart devices and services, but this model is increasingly common among Norwegian

Table 1 Overview of analytical methods employed

Method	Purpose	Contribution
Word Analysis (TF-IDF)	Identifies key terms by measuring their frequency within individual reviews and across the dataset	Highlights specific user concerns or interests, forming the basis for deeper sentiment and topic analyses
Sentiment Analysis (RoBERTa)	Classifies user sentiment (positive, neutral, negative) using advanced contextual interpretation, including sarcasm and nuanced language	Gauges overall sentiment in user feedback, complementing word analysis to create a balanced understanding of user perceptions
Topic modelling (LDA)	Extracts prominent topics by analyzing word co-occurrence patterns and uncovering thematic relationships	Identifies key themes and user concerns, informing more targeted qualitative analysis and enhancing interpretative accuracy
Exploratory Qualitative Analysis	Applies thematic analysis through manual coding, filtering reviews for relevance based on keywords	Provides nuanced insights into user perspectives, highlighting the most relevant comments and complementing automated findings

energy companies, making Tibber a representative example of a broader market trend.

App features overview

The app offers advanced electricity consumption details and personalised analytics through an interface with interactive icons and widgets (Tibber, 2024). The app's *Home* tab offers a variety of choices, including *Electricity Price*, *Consumption*, *Outdoor temperature*, and an *Add more* options. The *Analysis* tab provides an in-depth insight into *consumption* (hourly, daily, monthly). Users can go through detailed graphs to see more about the hourly electricity consumption in krone or kWh and its correlation with electricity price or the outside temperature. The *Compared to others* option allows users to compare their electricity use with similar households based on factors such as electric vehicle use, heating methods and household size. Using machine learning and data analysis, the *Consumed the Most* feature provides a share distribution revealing how energy is consumed (i.e., heating, always-on appliances, behaviour, car charging). In addition, the *Power-ups* option enables the users to integrate additional features such as real-time tracking, smart charging, smart heating, solar production, smart home integration and others. In addition to these main feedback screens, the app offers push notifications, service updates, and personalised energy usage highlights. Additionally, the app uses machine learning and artificial intelligence to learn household energy consumption patterns and provide more personalised feedback on consumption distribution (Tibber, 2023), which could be effective for behaviour change (Sardianos et al., 2021).

Data collection process

User reviews were gathered from both the Google Play Store and the App Store, specifically from Norwegian users, to ensure relevance to our research scope. The online user reviews were exported using Python programming language and the web scraping Python modules: *google-play-scraper (*https://pypi.org/project/google-play-scraper/*)* and *app-store-scraper (*https://pypi.org/project/app-store-scraper/*),* which allowed us to export the datasets from the Android as well as iOS platform. We limited the reviews to Norway as per our scope. We collected a total of 3603 reviews, including 941 from the App Store (posted between 2017–10–24 to 2023–02–20) and 2662 reviews from the Google Play Store (posted between 2017–02–16 to 2023–02–20). This collection was conducted in February 2023. Each review record contains attributes such as source, reviewer ID, username, title, description, rating, and date.

Data pre-processing

We transformed raw data into a structured format to facilitate broader accessibility and understanding. Several pre-processing steps were undertaken to transform the raw data into an understandable structure.

After scraping the reviews from both platforms, we compiled them into a single data frame. The reviews were originally written in Norwegian. While recent advancements in Natural Language Processing (NLP) tools such as BERT allow for direct analysis of texts in their original language (Kummervold et al., 2021; Vågerö et al., 2023), reviews were translated into English. This decision was made to align with the broader English-speaking audience, thus easing the dissemination of the research. Translating the reviews ensured both data validity and result replicability. The translations were performed using the Google Translate function in Google Sheets using the formula (=GOOGLETRANSLATE (A1, "no", "en")). This enabled efficient and immediate translation for our large datasets of over 3,000 reviews. After translation, we proceeded with additional pre-processing steps, including removing duplicates, punctuations and stop words, lowercasing the text, tokenisation, and lemmatisation (Abdelrahman et al., 2021; Huang et al., 2022; Serrano et al., 2021).

Dataset overview

Figure 1 presents the temporal distribution of reviews on Android and iOS platforms, which shows a growing number of user reviews across both platforms, suggesting increasing app engagement. The seasonal peaks suggest increased engagement during winter. A similar pattern in review activity is observed across both platforms over time, highlighting user engagement regardless of their operating system preference.

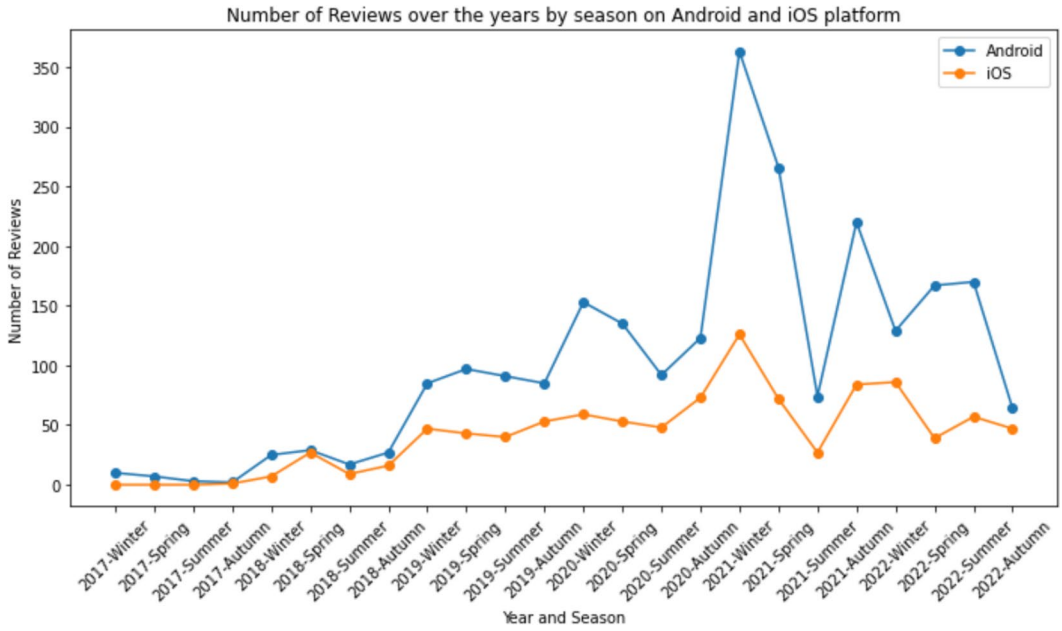


Fig. 1 Review trends over the years by season on Android and iOS platforms

Figure 2 presents the temporal review distribution against the dynamic electricity price market calculated as the arithmetic average of prices in the five zones (NVE-RME, 2024), with standard deviation indicating price variability. This data overview sets the stage for analysing how price fluctuations and seasonal changes might impact app usage and review patterns. Significant price fluctuations began in Autumn 2021 and intensified with the escalation of the Russia-Ukraine conflict in February 2022, profoundly affecting global energy and commodity markets. Given that Norway is a significant energy exporter, these fluctuations were notably reflected in the Norwegian market (energifakta, 2022). We can observe that user reviews started to rise in 2019. Even though the price appears to be lower in 2020, the number of reviews continued to increase. Additionally, seasonal effects were observed, as there were consumption spikes in winter. This may indicate increased user attention to energy management during periods of higher usage and potentially higher costs, prompting them to leave reviews.

This dataset overview establishes an initial understanding of how user review patterns evolve over time, setting the stage for more detailed analyses of how digital energy platforms like Tibber mediate user interactions with energy insights.

Analytical framework

We utilise both quantitative and qualitative techniques leveraging established natural language processing (NLP) tools and machine learning algorithms to analyse user-generated feedback on the Tibber app. This mixed-methods approach enables us to uncover insights into user interactions with digital energy platforms and identify key themes in their engagement. We aim to uncover user perceptions, motivations, and the app's mediating role in shaping digital energy engagements by integrating methods such as word frequency analysis, sentiment analysis, topic modelling, and exploratory qualitative analysis. Figure 3 presents the analytical framework guiding this study, outlining how various techniques are combined to extract insights from app reviews.

The subsequent Sects. (3.3.1 to 3.3.4) detail the methodologies and tools used in each stage of our analysis, explaining how they contribute to our research objectives and the insights they are expected to yield.

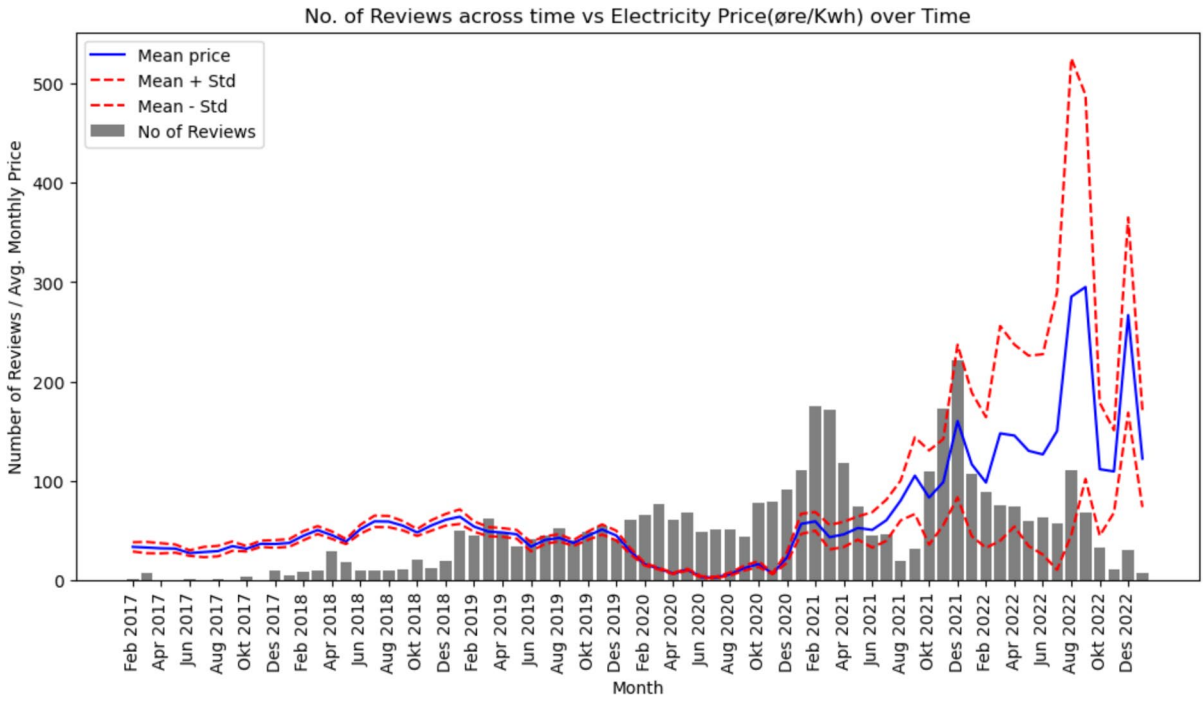


Fig. 2 No of Reviews vs Electricity price over time

We extensively used Python programming language and libraries within the Python 3.9 version. In addition, libraries such as NumPy, Pandas, NLTK, seaborn and LDA, among others, were used during the data processing pipeline for various data-specific tasks such as cleaning, aggregating, and visualisation.

Word analysis

We employed the TF-IDF (Term Frequency-Inverse Document Frequency) approach to identify key terms and expressions characterising user reviews. Unlike simple frequency counts, TF-IDF assesses the relative importance of words within individual reviews compared to the entire dataset, enabling us to highlight distinctive and contextually significant terms (Mishra et al., 2019; Qaiser & Ali, 2018). It combines two metrics: term frequency (TF), which indicates how frequently a term appears in a document, with inverse document frequency (IDF), which assesses the significance of a word across the entire dataset. TF-IDF value increases with word frequency within a document but is adjusted downward based on the overall prevalence, highlighting terms more distinctive to a particular document (Ramos, 2003).

This approach allowed us to identify keywords that are frequent within individual reviews and significant compared to their presence across the entire dataset. The output of this analysis included word clouds and bar charts, revealing the most prominent unigrams, bigrams, and trigrams within the user reviews.

Sentiment analysis

Sentiment analysis is the process of characterising and sorting people's opinions, sentiments, and attitudes conveyed in a piece of user-generated text (i.e., reviews, online feedback and social media posts) to determine their perception (positive, negative or neutral) towards a particular product or service (Medhat et al., 2014; Zhang & Liu, 2017). It aims to comprehend and forecast social phenomena by evaluating people's attitudes and moods (Pozzi et al., 2017).

We utilised Natural Language Processing (NLP) techniques to conduct sentiment analysis, the typical method for processing and analysing a large amount of textual data. Specifically, we applied the

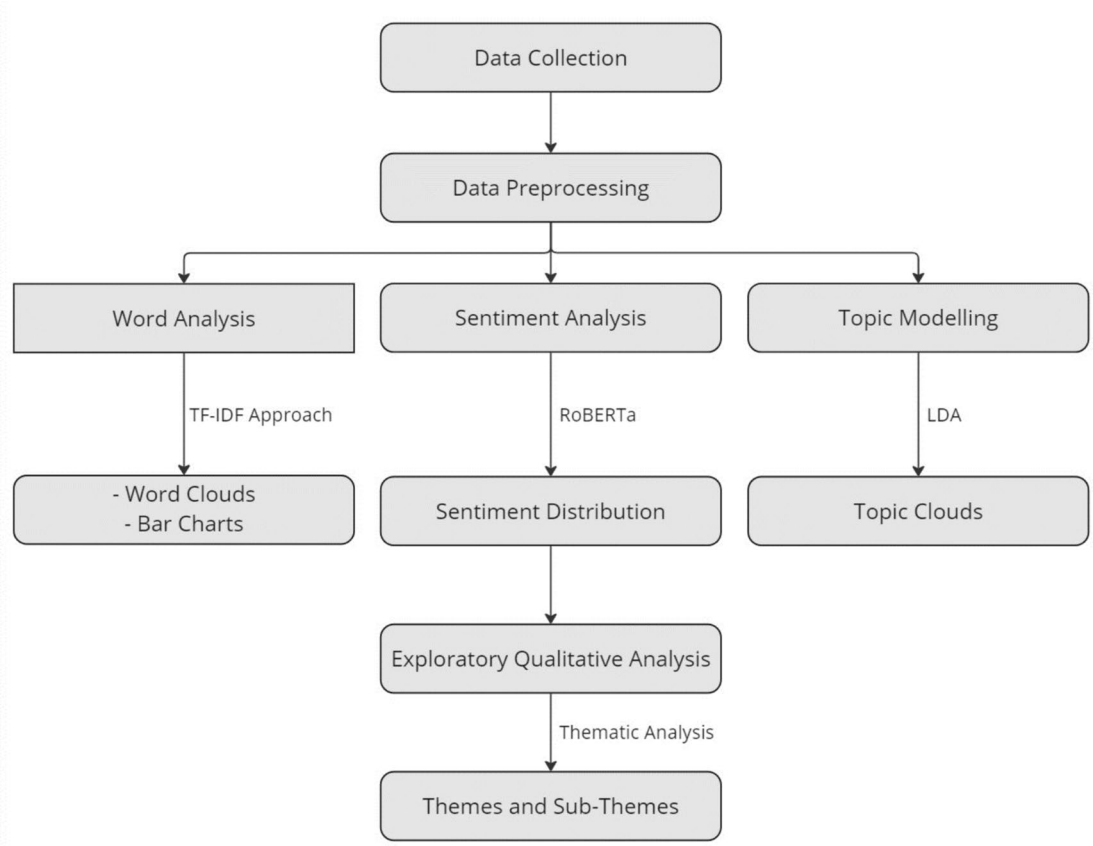


Fig. 3 Analytical Framework

Robustly Optimised BERT Approach (RoBERTa), a pre-trained language model developed by the Meta Research team (Liu et al., 2019) based on the Bidirectional Encoder Representations from the Transformers (BERT) method (Devlin et al., 2018; D. Q. Nguyen et al., 2020a, 2020b). RoBERTa reads and understands the entire review, considering the context and nuances of the language used, and transforms this understanding into a numerical representation. This representation is fed into a classification system trained on many examples of positive, negative, and neutral sentiments. The system then decides which category the review falls into based on its overall content and tone, not just individual words (Q. T. Nguyen et al., 2020a, 2020b). This method's key advantage is its ability to identify complex linguistic patterns, sarcasm, and context-dependent opinions, providing a more accurate sentiment analysis than simpler keyword-based approaches (Q. T. Nguyen et al., 2020a, 2020b).

Topic modelling analysis

Topic modelling is a widespread machine-learning technique that has recently gained popularity across various domains. It relies on statistical modelling to extract topical patterns within a collection of texts and thus automatically deduces semantic topics from a text corpus (Kherwa & Bansal, 2019). Our study utilised a Latent Dirichlet Allocation (LDA) based topic model; a generative probabilistic model founded on a three-level hierarchical Bayesian model (Blei et al., 2003). This method facilitated the identification of prevalent themes among user reviews, providing insights into users' perceptions, motivations, and interactions.

Our approach was primarily inductive, allowing themes to emerge organically from the data. However, we specifically focused on topics tied to usage motivations, user perceptions of the app's functionalities, and its potential as a digital tool to support sustainable energy practices. The LDA model revealed clusters of frequently co-occurring words, which we interpreted in this context.

The coherence measure was used to identify the optimal number of topics, complemented by manual evaluation to ensure a comprehensive topic quality assessment (Mimno et al., 2011; Röder et al., 2015). Four topics were chosen for exploration, deriving labels based on interpreting the most frequent words. In LDA topic modelling, the proximity between topics helps understand their similarities and relationships, aiding in interpretation, improving model quality, and facilitating more effective document analysis and content recommendations (Blei et al., 2003).

Exploratory qualitative analysis

To complement the large-scale, automated insights generated through topic modelling, we conducted an exploratory qualitative analysis using thematic analysis. This approach emphasizes detailed human interpretation, enabling a deeper understanding of user experiences and perceptions that automated methods alone could not capture (Gillies et al., 2022).

Using Python, we filtered reviews based on carefully selected keywords associated with awareness, motivation, engagement and potential behavioural adjustment. While the reviews were not explicitly designed to measure behaviour, terms such as '*motivation*', '*habit*', '*routine*', '*aware*', '*track*', '*feedback*', '*progress*', '*follow*', '*commit*', '*improve*', '*adapt*', '*learn*', '*explore*', '*discover*', '*engage*', '*intention*', '*environment*', '*sustainable*' and '*green*', were chosen to identify user-reported experiences that indirectly reflect these ideas.

The analysis followed the six-phase thematic analysis framework Braun and Clarke (2006) outlined. This process included familiarizing with the data, generating initial codes, identifying themes, refining themes, defining and naming themes, and creating a report. Through this structured approach, we extracted themes and sub-themes from the filtered reviews, shedding light on users' motivations, perceptions, and interactions with the app.

Although this approach may have excluded some relevant reviews not containing these specific keywords, and it might have limited fully capturing the user experiences of the wider user base, it allows for a more targeted analysis of the most relevant comments, potentially uncovering patterns and themes that may be hidden in more general analyses.

Analysis and findings

Table 2 summarizes the analytical methods used, highlighting their primary purpose and additional contributions to enhancing the overall understanding of user feedback.

Word analysis

Figure 4 presents a word cloud visualisation where larger font sizes indicate words with higher TF-IDF scores in the user reviews dataset. Figure 5 shows the top-scoring bigrams and trigrams. The analysis revealed that users frequently mentioned the app itself (*good app*, *clear app*, *super app*), highlighting its positive points (*good*, *clear*, *simple*, *easy*). Additionally, users emphasised their experience with the app's functionality (*works well*, *easy use*, *full control*), particularly regarding monitoring (*overview*, *consumption*, *power*) and managing (*control*, *use, follow*) their electricity use. Notably, users praised the app's clarity (*clear*, *simple*, *straightforward*) and user-friendliness (*user friendly*, *easy use*), underscoring its effectiveness as a digital tool that meets users' energy management needs. As illustrated in Figures 4 and 5, these insights highlight the app's key strengths and provide an overview of user engagement and satisfaction.

Analysing both bigrams and trigrams allowed us to capture varying levels of granularity in user feedback. Bigrams provided high-level insights into frequently mentioned terms, while trigrams added richer context by highlighting specific functionalities or experiences users value.

Sentiment analysis

Figure 6 presents the overall sentiment distribution, revealing that reviews predominantly indicate positive sentiments, accompanied by a significant number of neutral and relatively few negative sentiments.

Table 2 Overview of analytical methods and their contributions

Method	Purpose	Contribution
Word Analysis (TF-IDF)	Identifies key terms by measuring their frequency within individual reviews and across the dataset	Highlights specific user concerns or interests, forming the basis for deeper sentiment and topic analyses
Sentiment Analysis (RoBERTa)	Classifies user sentiment (positive, neutral, negative) using advanced contextual interpretation, including sarcasm and nuanced language	Gauges overall sentiment in user feedback, complementing word analysis to create a balanced understanding of user perceptions
Topic modelling (LDA)	Extracts prominent topics by analyzing word co-occurrence patterns and uncovering thematic relationships	Identifies key themes and user concerns, informing more targeted qualitative analysis and enhancing interpretative accuracy
Exploratory Qualitative Analysis	Applies thematic analysis through manual coding, filtering reviews for relevance based on keywords	Provides nuanced insights into user perspectives, highlighting the most relevant comments and complementing automated findings

Additionally, we examined the app ratings provided by users. Figure 7 correlates these ratings with the sentiment categories, showing that users with positive or neutral sentiments frequently awarded the app a 5-star rating, suggesting that they perceive the app as a valuable digital solution.

Despite the overall positive perception, word cloud was generated from the small number of negative reviews (182 out of 3331), as presented in Fig. 8. The word cloud highlights app performance issues, pricing and billing concerns, customer service challenges, and core functionality problems, providing a clear roadmap for targeted improvements.

To address our research questions, we focused on positive and neutral reviews to explore user engagement with the app, its connection to motivation, and any potential influence on household practices and behavioural adjustments. This approach positions user engagement as the central focus and examines its potential to support sustainable household practices.

To understand how users perceive various aspects of the app, we analysed the most common parts of speech (nouns, verbs, adjectives, and adverbs) used in the reviews, as shown in Fig. 9. We found nouns referring to energy-related functionalities of the app and the feedback it offers. The verbs refer to the energy practices the users are actively engaged in. The adjectives highlighted the users' view towards the energy use feedback or the app as an easy and useful solution, considering it a user-friendly tool. The presence of emphatic adverbs such as *absolutely* and *incredibly* could indicate high levels of user satisfaction, but additional analysis is necessary to understand exact intentions.

Topic modelling analysis

We employed the Latent Dirichlet Allocation (LDA) technique for topic modelling to gain deeper insights into positive and neutral reviews. Given the relatively high number of these reviews, a comprehensive understanding of user satisfaction and app strengths was ensured. This approach led to more accurate and actionable insights. As the number of negative reviews is low, we excluded them from topic modelling and thematic analysis, as they did not provide a reliable basis for analysis. This strategy enabled us to effectively uncover key themes and guide future improvements, leveraging the most substantial data for a comprehensive analysis.



Fig. 4 Word Cloud Based on TF-IDF approach

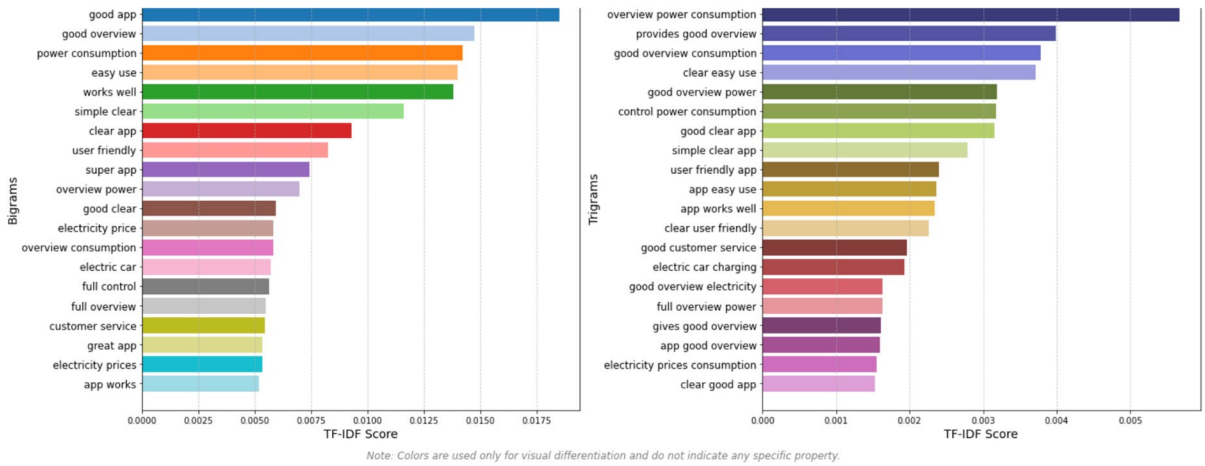


Fig. 5 Top-scoring bigrams and trigrams based on the TF-IDF approach

We selected four topics for exploration based on a combination of coherence measure, interpretability and alignment with the research questions. The resulting topics were custom-labelled as themes based on the researcher's interpretation of the most frequent words.

Figure 10 displays the word clouds of the most important top occurring words clustered under the four distinctive topics, Table 3 outlines the four topics identified based on keyword clustering and interpretation, while Fig. 11 depicts the LDA model's inter-topic distance map as presented by (Chuang et al., 2012; Sievert & Shirley, 2014).

Topic 1: Smart Energy Consumption This topic emerged as the most discussed, highlighting user engagement with energy-saving features for cost management. It suggests that users particularly value real-time energy usage insights, empowering them to make informed decisions to reduce consumption during peak hours and optimize the use of household appliances.

Topic 2: Electric vehicle charging Is the second most popular topic. It highlights user engagement with the app's smart EV charging functionality, emphasizing device compatibility and the integration of

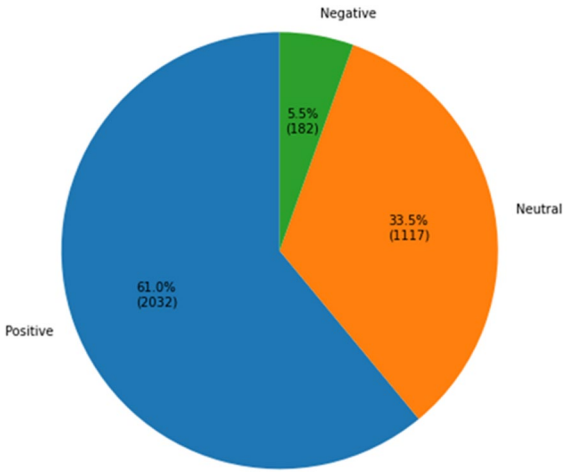


Fig. 6 Sentiment distribution of the reviews

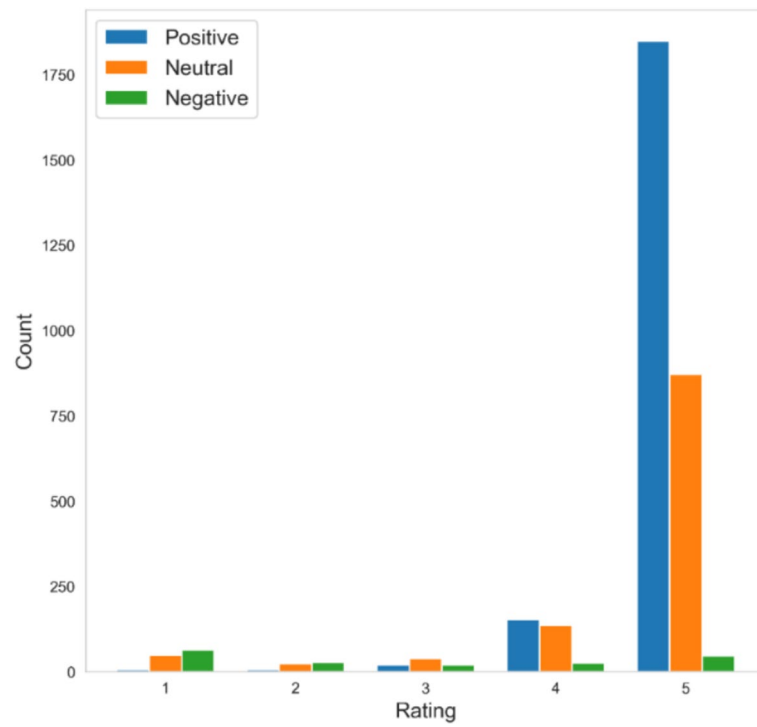
innovative energy management paradigms. Additionally, keyword clustering suggests that the app's features could extend beyond EV charging to interconnect with other IoT-enabled household devices, such as heat pumps and smart appliances, supporting a more integrated approach to home energy management.

Topic 3: User-friendliness This topic underscores the value of clear, intuitive app design and the availability of actionable information in promoting user engagement and highlights that users tend to prefer apps that simplify complex energy management tasks, minimize learning barriers, and enhance the overall user experience.

Topic 4: Overall Satisfaction This topic reveals that positive experiences with smart energy apps significantly boost overall satisfaction and sustained engagement and further highlights the app's ability to meet diverse household energy management needs, such as cost savings, convenience and trust among users.

In our topic modelling results, as highlighted by the inter-topic distance map and the identified themes, Topics 1 and 2 relate primarily to technological aspects, with Topic 1 on energy efficiency and cost-reduction approaches and Topic 2 on integrating electric vehicles with smart energy systems. Both topics underscore the importance of tools that enable users to manage energy consumption effectively. Topics 3 and 4, in contrast, highlight user experience, with usability and satisfaction as key engagement drivers.

Fig. 7 Sentiment vs Rating



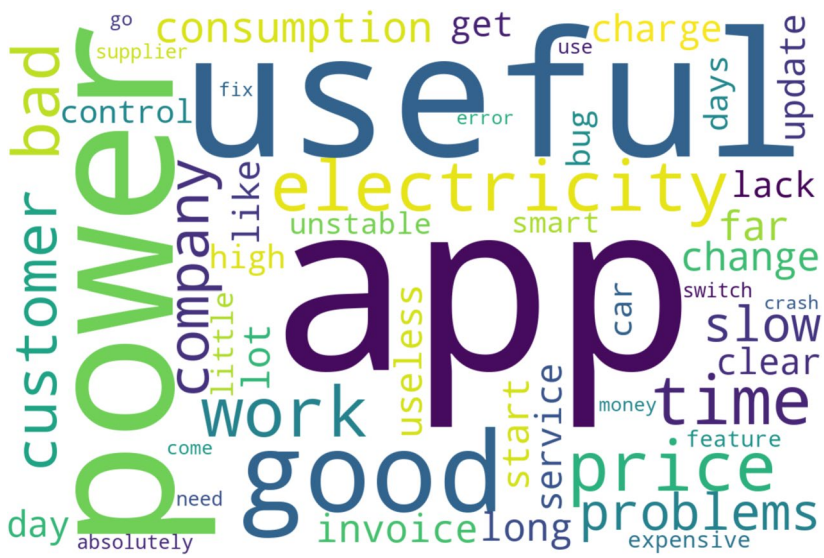


Fig. 8 Word Cloud for negative reviews

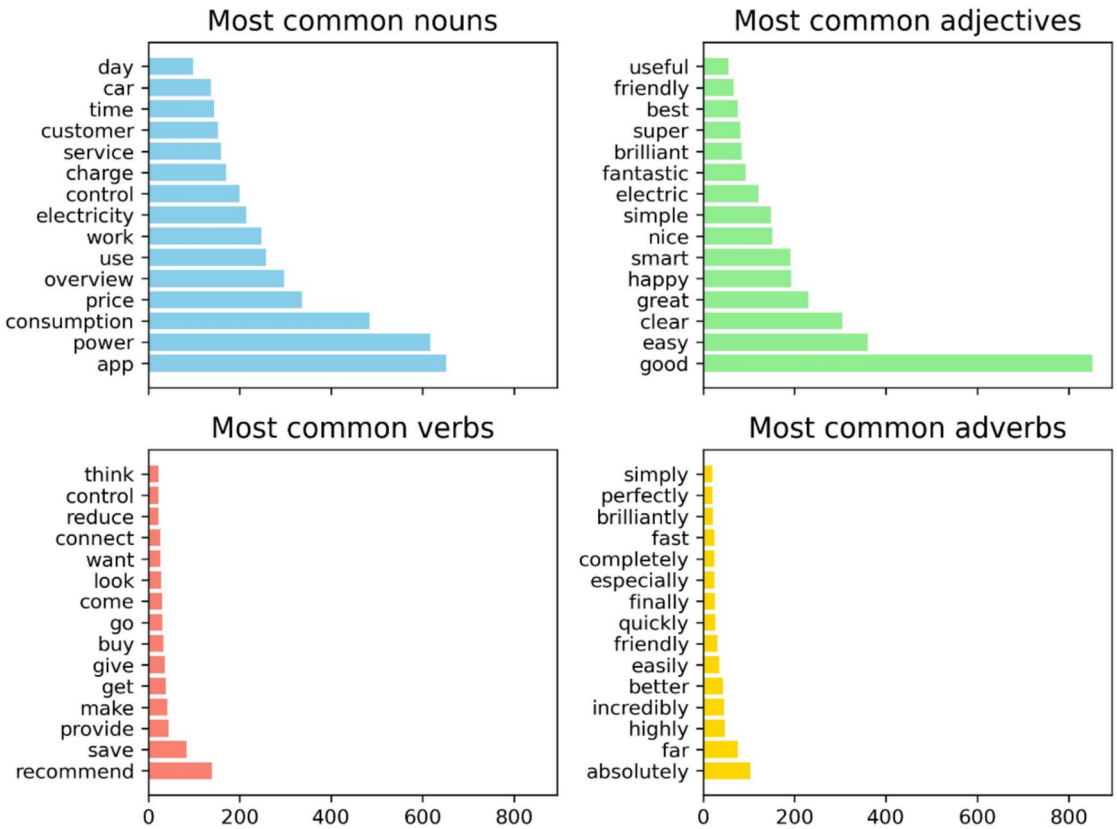


Fig. 9 Frequency analysis of parts of speech within the positive and neutral reviews

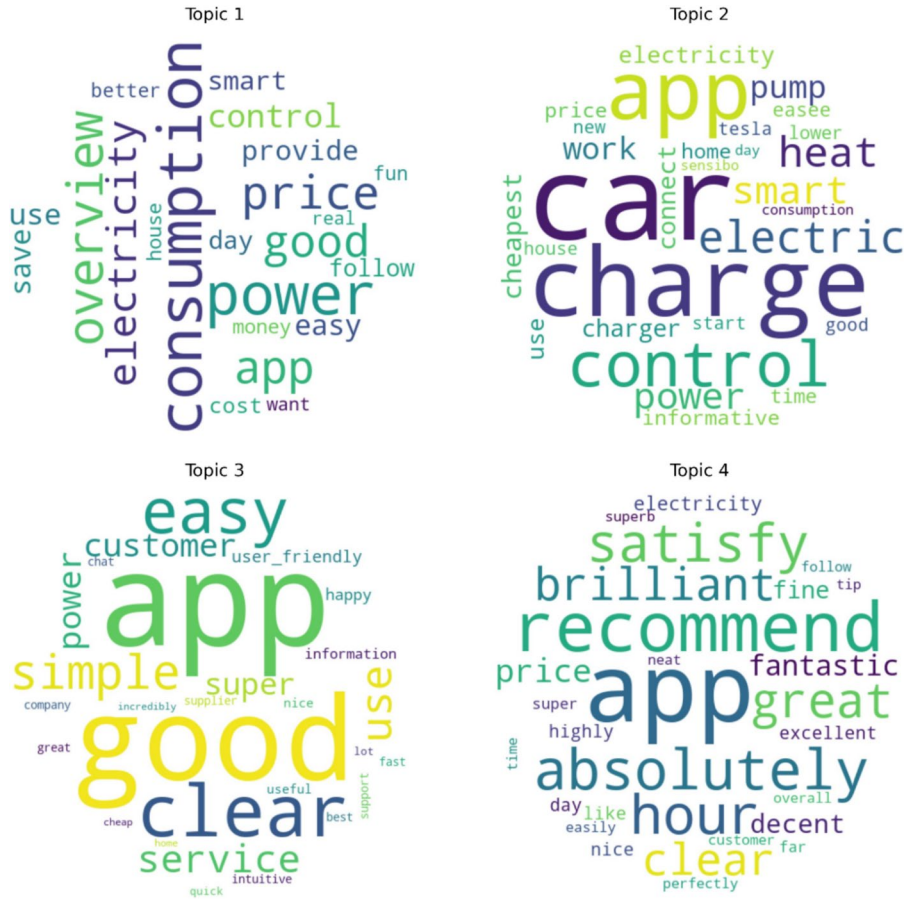


Fig. 10 Word clouds displaying the four topics' most important words

Table 3 Resulting Topics based on the LDA model

Topic	Theme	Feature words
1	Smart energy consumption	*consumption, power, overview, price, app, good, electricity, control, use, save, easy, provide, smart, day, follow, cost, house, money, fun, better, real, want*
2	Electric Vehicle Charging	*charge, car, app, control, electric, heat, smart, power, pump, work, cheapest, cars, connect, charger, use, electricity, informative, price, home, time, easee, lower, house, tesla, start, new, good, consumption, sensibo, day*
3	User-Friendliness	*app, good, clear, easy, simple, use, service, power, customer, super, user_friendly, happy, information, nice, useful, supplier, company, best, intuitive, great, support, chat, fast, quick, lot, cheap, home, incredibly*
4	Overall Satisfaction	*app, recommend, absolutely, hour, satisfy, great, brilliant, clear, price, fantastic, decent, fine, electricity, day, like, excellent, highly, nice, super, far, overall, neat, tip, perfectly, easily, time, follow, superb, customer*

Users use the app mainly for cost management, convenience, and an intuitive interface. The app's balance between functionality and user-centric design supports personalized energy management while contributing to sustainability goals.

Exploratory qualitative analysis

Based on the screening criteria, 279 reviews were included in the analysis. Table 4 presents user reviews' themes, subthemes, and illustrative examples.

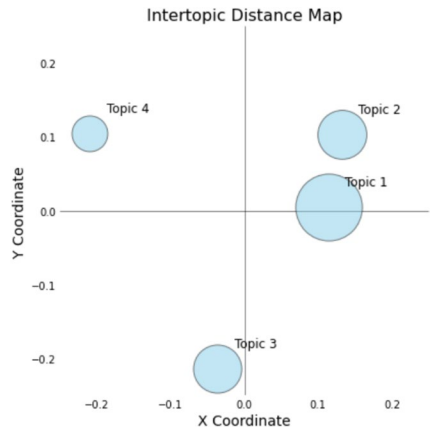


Fig. 11 Inter-topic Distance Map

Table 4 highlights that the app is a powerful awareness tool, offering innovative visualisations that enhance users' understanding of their energy consumption patterns and electricity prices. Its integration with smart home devices and electric vehicles demonstrates advanced smart energy management capabilities. The app successfully motivates users by providing incentives to switch suppliers, fostering energy consciousness, and enabling social comparisons. Users demonstrate a potential interest in energy efficiency including consumption optimisation and reduction, conscious energy use and adopting more sustainable technologies, with a focus on potential future upgrades. The app's user-friendly interface, clear information presentation, and responsive customer support enhance user experience. Additionally, the focus on environmental sustainability through renewable energy sources, sustainable consumption practices, and in-app green product offerings aligns with users' growing interest in eco-friendly solutions. The app effectively combines energy feedback, smart home integration, user engagement, and sustainability features that promote energy efficiency and conservation.

Discussions

By analysing a comprehensive dataset of user reviews from a smart energy app in the Norwegian context, this research provides novel insights into the dynamics of user engagement, perceptions, and energy management practices. The findings reveal that user engagement with digital energy platforms is shaped by a combination of intrinsic motivators, functional utility, and emotional gratifications, offering significant implications for designing user-centric energy solutions.

A central theme emerging from the findings is that user engagement with smart energy apps is not solely driven by financial incentives, challenging the prevailing assumption that cost-saving incentives are the primary drivers of adoption. Instead, intrinsic motivators such as control, empowerment and convenience play a critical role in sustaining engagement. This aligns with Stern (2000)'s assertion that environmentally beneficial actions often arise from non-environmental motivations, such as personal competence or social relationships. The findings suggest that digital energy platforms must cater to these intrinsic motivators to foster long-term engagement.

Word and sentiment analyses reveal that users value the app's intuitive design, seamless usability, and energy optimization features. Frequently mentioned themes, such as "overview power consumption," "electric car charging," and "user-friendly interface," highlight the app's appeal to a diverse range of users. These findings underscore the importance of designing platforms that balance functional utility with emotional gratifications. The positive sentiment associated with features like real-time monitoring, cost optimization, and gamification further emphasizes the role of hedonic quality in technology adoption, as Ferreira et al. (2023) noted.

Topic modelling and thematic analysis provide deeper insights into the specific functionalities that drive engagement. Users engage most with features related to smart energy consumption and EV charging, reflecting Norway's advanced adoption of electric vehicles (Schulz & Rode, 2022). However, the limited reference to solar photovoltaic systems aligns with their low uptake in Norwegian residential settings (Inderberg et al., 2016). These findings highlight the necessity of tailoring digital energy platforms to regional technological contexts and user practices to ensure broader adoption and sustained utilization.

Implications for designing user-centric digital energy solutions

The study's findings have significant implications designing user-centric digital energy solutions. First,

Table 4. Thematic Analysis

Themes	Sub-themes	Representative Example
Energy Feedback and Awareness	Awareness tool	*"A nice tool that helps to aware you on power consumption and (less good) habits."*
	Innovative visualisation	*"An app that provides informative logical info by motivating and engaging its customers to follow in the future. Fantastic innovative solution with high customer focus."*
	Better Understanding	*"Incredibly good to be able to follow the electricity price every day and plan my power consumption based on it!"*
Smart Home and Device Integration	Electric vehicle integration	*"The charge of our Tesla is controlled by the app, so everything works as promised by tibber."*
	Smart home device compatibility	*"Has Smart House from Homey and the Tibber app fits perfectly with this one."*
	Smart energy management	*"Much has been automated with both charging and heating."*
User Motivation and Engagement	Incentive to switch supplier	*"We switched to Tibber in 2019 and love it. Cheap power, lovely app with superb features, and perfect for us as a slower with some smart home solutions."*
	Energy consciousness	*"Tibber makes it easy, accessible and fun to follow and control their own power consumption."*
	Social comparison features	*"And then I can keep an eye on consumption and see how consumption I have in relation to others, and whether my consumption goes up or down."*
	Daily engagement	*"I am on the app daily and follow my consumption and can put my electric car on charging, washing machine, dryer, swimming/showering etc."*
Energy efficiency	Consumption optimization and reduction	*"The app has made me more aware of my power consumption and thus meant that I have reduced consumption by approx. 15% After optimisation of time for night lowering and temperatures 👍"*
	Conscious energy use	*"Very nice to know at all times what the current costs. Then I particularly adapt to laundry and dryer at the best prices. 👍💯"*
	Potential Future upgrade	*"Looking forward to connecting electric car charges and maybe a day solar cell :)"*
App Usability and User Experience	Intuitive interface	*"Simple and clear app. It is very simple. Maybe a little too simple for me to understand."*
	Consumption and price tracking	*"Incredibly good to be able to follow the electricity price every day and plan my power consumption based on it!"*
	Information clarity	*"Clear and easy to follow :)"*
	Customer support	*"Very good customer service and information along the way.👍"*
Environmental Sustainability	Renewable energy sources	*"Environmentally friendly too. You get details about where you get power from."*
	Sustainable consumption practices	*"Exciting and innovative app that allows you to control the energy of your house smoother."*
	In-app green product offerings	*"Tibber sells / has solutions for various savings measures."*

the app's potential to promote sustainable practices through innovative visualization tools, user engagement, smart home integration, and heightened energy awareness positions it as a promising mediating tool for behavioural adjustments. Users report enhanced awareness of their energy consumption and optimized usage patterns, suggesting that digital energy platforms can support the broader paradigm of consumer participation in energy transition, contributing to energy democratization (Wemyss et al., 2023).

Second, the app's transparency and ease of use, alongside specialized options such as EV charging and energy management, cater to a wide range of user preferences. These findings underscore the importance of designing platforms that are both accessible and adaptable to diverse user needs. Feedback from negative reviews, highlighting usability problems and unmet expectations, provides practical recommendations for developers to improve user satisfaction and address barriers to engagement.

Third, the study extends existing research beyond pilot-based studies and small-scale user surveys. Unlike prior work that relied on limited, self-reported data, this analysis incorporates a large dataset of active users, offering generalizable insights into real-world engagement with digital energy solutions. The findings highlight the dual role of these platforms in providing functional utility and facilitating potential behavioural adjustments, empowering users to make informed decisions and contribute to sustainable energy consumption practices.

Contribution to the existing knowledge

This research contributes to the evolving discourse on digital energy solutions by integrating insights from sociology, technology studies, data science, and behavioural science. Sociology illuminates how social and cultural contexts influence app usage and user preferences, while technology studies emphasize how specific app features can foster engagement and energy-conscious practices. Data science facilitates the processing and analysis of large datasets, uncovering patterns in user sentiments and interactions, and behavioural science explores the intrinsic motivations that drive user engagement, such as empowerment, convenience, and personal competence. Unlike previous studies that predominantly relied on pilot projects and lacked population-level data, this research utilises an extensive dataset from active users, addressing the significant gaps in the literature on app-based digital energy feedback. Moreover, this research emphasizes the importance of user-centred features in encouraging engagement with digital energy services (Geelen et al., 2019; Westskog et al., 2015; Xun et al., 2023).

This study provides a nuanced understanding of how regional factors shape user engagement by situating these findings within Norway's specific energy context, marked by widespread smartphone use, enhanced digital skills, and growing public interest in managing household energy consumption (Paneru & Tarigan, 2023). The interdisciplinary and empirically grounded approach adopted in this research informs the development of more user-centric and context-sensitive energy management tools, aligning with diverse user needs and advancing sustainable energy practices.

Limitations and further research

We acknowledge several limitations inherent in this study and identify opportunities for future research to build on its findings. Engagement with the app does not guarantee measurable energy savings, and future research should incorporate empirical analysis of electricity consumption data to confirm reductions or shifts in peak usage. Additionally, the study's focus on the Norwegian context may limit the generalizability of the findings to other regions with different energy landscapes and user practices. Future research should explore cross-cultural comparisons to identify universal design principles for digital energy platforms.

While focusing on energy engagement might have limited the scope, this study provides valuable insights into user interactions with energy feedback and the app's potential role in supporting energy practices. These results are crucial for advancing future energy efficiency strategies and refining app design.

By focusing on positive and neutral reviews, we ensured a robust dataset for analysis. Although the number of negative reviews was relatively low, their exclusion likely did not significantly impact the findings. To account for this, we included a word cloud analysis of negative reviews, highlighting key areas of dissatisfaction. As the app's user base expands

and the volume of reviews increases, future research could leverage a more substantial dataset of negative reviews, providing a broader understanding of user experiences and identifying additional opportunities for improvement.

The exclusion of reviews not containing specific keywords in the exploratory qualitative analysis may have limited our ability to capture the experiences of the wider user base fully. However, this targeted approach facilitated the concentrated analysis of the most pertinent comments, identifying patterns and themes that might have been obscured in a broader analysis.

Future research can address these limitations through a multifaceted approach, including thoroughly analysing negative reviews once their volume is sufficient for meaningful insights, leveraging alternative data sources beyond app store reviews, and integrating demographic data to inform targeted interventions. Comparative analyses across multiple energy apps could yield broader applicability, while longitudinal studies incorporating household-level energy consumption data and control groups would offer more robust evaluations of app impact. Supplementing app review analysis with qualitative methodologies like surveys and interviews would further reveal user motivations and experiences.

Conclusion

This research demonstrates a systematic approach to transforming user feedback into actionable insights, highlighting smart energy apps as promising digital energy engagement tools. These insights aim to support energy efficiency initiatives. The study focuses on how Norwegian users engage with a smart energy app, analysing their interactions, perceptions, motivations, and the app's influence on energy management.

The findings reveal frequent use of features such as monitoring energy consumption, managing appliances, and integrating smart home technologies like EV charging, which directly contribute to improved energy efficiency. Users express predominantly positive feedback, highlighting the app's intuitive design, real-time feedback, and gamification features, all of which enhance usability. Motivations for sustained app use extend beyond cost savings to include empowerment, convenience, and alignment with sustainability values. Additionally, the app has a notable influence on energy management, fostering increased energy awareness, supporting energy efficiency initiatives, and promoting ease of use and environmental responsibility.

These findings position the app as a practical tool for energy management and a mediator of sustainable practices. While this study focuses on the Norwegian context, its insights extend beyond national boundaries. Norway's advanced smart grid infrastructure and high digital literacy make it a relevant case for understanding digital energy engagement. The findings contribute to broader discussions on energy digitalization, offering valuable lessons for other countries with similar regulatory frameworks and technological readiness. Moreover, they provide a reference point for nations aiming to achieve similar advancements in digital energy solutions by leveraging user-centered design and policy-driven innovation.

By integrating interdisciplinary insights, this study provides a comprehensive framework for understanding digital energy engagement and guiding the development of user-centred energy solutions. Despite some limitations, this research advances the smart and sustainable energy field by analysing user-generated content to uncover motivations and preferences. The findings offer actionable insights for improving app design and informing policies that support sustainable energy transitions.

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Author contributions Chandra Prakash Paneru: Conceptualisation, Methodology, Programming, Writing—original draft, review & editing, Project administration. Ari K. M. Tarigan: Resources, Writing—review & editing, Supervision. Cristian Tosa: Guidance—Methodology, Programming and reviewing.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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